ACQ Data Architecture Overview

Data Acquisition layer within the enterprise Data Lakehouse

Vendor architecture review | Prepared from Proposed ACQ Data Architecture (January 2022)

Executive summary. The ACQ layer is the controlled acquisition and landing foundation for client, external reference, and internal application data. It preserves source fidelity, enforces Company/Division/Client segregation, applies consistent structures and naming, and prepares trusted data for downstream presentation, analytics, reporting, and AI workloads.

Where ACQ Fits in the Lakehouse

SOURCE SYSTEMS	AZURE BLOB STORAGE	ACQ LAYER	PRESENTATION	CONSUMPTION
Clients, PBMs, vendors, internal applications	Raw source files and business-aligned folders	Landing, typing, partitioning, history, deduplication	Curated dimensions and facts	Reporting, analytics, data science and AI

Core Design Decisions

1. Multi-tenant organization

The controlling hierarchy is Company -> Division -> Client. Data with the same format is co-located in common physical tables and partitioned by those business keys. Division-level sensitivity is preserved; where a client participates in multiple divisions, ACQ retains separate division-specific copies.

2. Three data domains

Client data includes claims, eligibility, and enrollment. External data includes shared vendor/reference sources such as Medi-Span, NCPDP, NPPES, and First DataBank. Internal data comes from applications such as PAX, AccuCast, and AMS360.

3. Single schema strategy

All acquisition tables use one database schema, DW_ACQ. Co-location reduces schema and security administration, simplifies downstream processing and prototyping, and makes acquisition data easier to discover.

4. Clear naming standards

Names are uppercase with underscores and descriptive words. The pattern identifies layer, data category, source, and client format, for example ACQ_CLIENT_PHARMACY_CLAIM_CVS, ACQ_EXTERNAL_MEDISPAN_AWP_PRICE, and ACQ_INTERNAL_PAX_ENROLLMENT.

Data Governance, Quality and Lineage

TYPE-1 / TYPE-2 HISTORY

Type-1 tables maintain one ACQ record per source business key or copy source-provided history as-is. Type-2 tables add warehouse-managed historical versions when prior values have analytical value. Type-2 tables use the _HISTORY suffix.

RAW + CURATED ACQ PATTERN

For sources requiring full source fidelity, an ACQ raw table captures every file row as strings, including duplicates and file metadata. The corresponding strongly typed ACQ table retains the selected active/master record for downstream use.

TRACEABILITY

A durable business key links duplicate transactions and supports navigation from presentation data back through ACQ to the original raw record and source file context.

CASE AND DATA TYPES

Source string values remain unchanged in ACQ and are standardized later between ACQ and presentation. The documented design retains case-insensitive SQL collation and uses non-Unicode string types to reduce space and improve ETL/query performance.

Azure Storage Organization

Data class	Organization	Purpose
Client	Container by Company/Division; folders by Client, Source, and Data Type	Supports security boundaries, discoverability, and client-specific operations
Multi-client file	Organized by sender/source rather than split by client	Preserves original file integrity, filename, receipt context, and metadata
External reference	Folders by vendor, such as First DataBank, Medi-Span, NCPDP, and NPPES	Supports shared reference data used across companies and divisions

Operational Approach

- Parallel modernization: the new data warehouse is built alongside existing legacy processes so functionality can migrate incrementally without disrupting current operations.
- File lifecycle: documented paths distinguish processing, archive, and skipped duplicate files, preserving operational context and supporting recovery and troubleshooting.
- Selective complexity: the two-table raw/curated pattern is used only where exact capture and duplicate resolution are required; simpler sources can load directly into ACQ.

Bottom line	ACQ provides the governed bridge between heterogeneous source data and curated lakehouse data products. Its value is secure segregation, consistent organization, source-to-report lineage, and a scalable onboarding pattern for additional clients, PBMs, vendors, and internal applications.